

Endterm Task 1: Issues and Limitations of GAD-NR

CS60078 Complex Network Theory | Spring 2025-26 | Project 3: Graph Anomalies

Group-3: 23EX10026, 22CS30072, 22IM30034

Paper: GAD-NR: Graph Anomaly Detection via Neighborhood Reconstruction (WSDM 2024)

1. Introduction

GAD-NR introduces neighbourhood reconstruction as a principled alternative to link-prediction-based graph autoencoders for anomaly detection. Our midterm analysis confirmed that the method outperforms 14 baselines on 5 of 6 datasets and is the only model that handles contextual, structural, and joint-type anomalies simultaneously. However, through careful reproduction, ablation, and quantitative analysis, we have identified several concrete issues and limitations. This report organises them into architectural, methodological, experimental, and scalability categories.

2. Architectural Limitations

2.1 One-Hop Neighbourhood Only

GAD-NR reconstructs only the immediate 1-hop neighbourhood of each node. The authors acknowledge this as a design choice for computational tractability, but it means the model is fundamentally blind to anomalies that manifest at larger structural scales. For example, a node that participates in an anomalous 2-hop or 3-hop motif — such as a bridge node connecting two otherwise-separated fraud rings — would be missed entirely. Extending to multi-hop reconstruction was mentioned as future work but not explored, and the $O(d)$ per-hop cost would compound to $O(d^L)$ for L hops, reintroducing scalability concerns.

2.2 Gaussian Approximation of Neighbour Distributions

The neighbourhood feature distribution is modelled as a single multivariate Gaussian. This is a strong unimodal assumption that breaks down when a node's neighbours are drawn from multiple distinct communities — a situation that is common in real-world social and citation networks. For instance, an interdisciplinary researcher connected to both biology and computer science clusters has a bimodal neighbour distribution. The Gaussian approximation would poorly fit such a node, potentially producing high reconstruction loss and triggering a false positive. The paper does not discuss this failure mode or evaluate on graphs with known multi-community structure.

2.3 Single-Layer GNN Encoder

The encoder uses a single GNN layer ($L=1$), which limits the receptive field to 1-hop. While this is consistent with the 1-hop decoder target, it means the node representations carry no information about broader graph context. Deeper encoders could improve representation quality, but the paper does not explore encoder depth beyond $L=1$, nor does it provide theoretical justification for why a single layer suffices for all anomaly types.

3. Methodological Issues

3.1 Anomaly Score Weighting is Heuristic

The final anomaly score is a weighted sum of three loss components: self-feature (λ_x), degree (λ_d), and neighbourhood (λ_n). While the paper claims robustness with fixed weights, the adaptive weighting scheme used in practice (normalising each loss by its range and applying weights $h=1.0$, $feature=2.0$, $degree=1.0$) is a heuristic with no theoretical grounding. In our reproduction, we observed that performance is sensitive to these inference-time weights — a poor choice can drop AUC by 10+ points on certain datasets. The paper does not provide a principled method for selecting these weights when the anomaly type is unknown.

3.2 Training Loss Scheduling is Ad Hoc

During training, the feature loss weight λ_2 is increased by 0.5 every *loss_step* epochs, while the degree loss weight λ_3 is halved. This schedule is not motivated by any theoretical analysis and introduces two additional hyperparameters (the step interval and the increment/decay rates). The paper does not ablate these scheduling choices, making it unclear whether the reported performance gains come from the neighbourhood reconstruction idea itself or from this particular scheduling trick.

3.3 Unsupervised Setting Assumes Anomaly Sparsity

Like all reconstruction-based anomaly detectors, GAD-NR relies on the assumption that anomalies are rare — the autoencoder learns to reconstruct "normal" patterns and flags high-loss nodes. When the anomaly ratio is high (e.g., Weibo at 10.3%), the model may partially learn anomalous patterns too, reducing discriminative power. Indeed, GAD-NR's performance on Weibo (87.7%) is lower than simpler methods like Radar (98.9%), suggesting the approach struggles when anomalies are not sufficiently sparse.

4. Experimental Limitations

4.1 Synthetic Anomaly Injection for Evaluation

For contextual, structural, and joint-type anomaly evaluation, the paper injects synthetic anomalies following the BOND protocol. These injected anomalies are simplistic by design: contextual anomalies are created by swapping features with the most distant node, and structural anomalies are formed by creating perfect cliques. Real-world anomalies are far subtler — a fraudster does not have the single most different feature profile, nor do bot clusters form perfect cliques. The strong performance on injected anomalies may therefore overstate GAD-NR's effectiveness on truly organic anomalies. Notably, on datasets with real anomaly labels (Weibo, Reddit, Enron), the absolute AUC values are much more modest.

4.2 Limited Dataset Diversity

All six datasets are static, undirected, and homogeneous. The paper does not evaluate on directed graphs (e.g., Twitter follow networks), heterogeneous graphs (multiple node/edge types), temporal/dynamic graphs (where anomalies evolve over time), or very large graphs (millions of nodes). Many real-world fraud detection scenarios involve precisely these graph types, so GAD-NR's applicability beyond the tested settings is unverified.

4.3 No Comparison with Recent Self-Supervised Methods

The baselines used in the paper are primarily from 2018-2022. More recent methods based on graph contrastive learning (e.g., ANEMONE, SL-GAD) and graph transformers have shown strong anomaly detection performance but are not included in the comparison. This makes it difficult to assess whether the neighbourhood reconstruction paradigm is truly superior to the latest state-of-the-art.

5. Scalability Concerns

While GAD-NR reduces per-node complexity from $O(d^3)$ (NWR-GAE) to $O(d)$, the overall training still requires iterating over all nodes, sampling neighbours, and computing KL divergence for each. On large graphs, the neighbour sampling loop (3 rounds per epoch) creates a bottleneck. In our CPU-based experiments, a single epoch on Cora (2,708 nodes) takes ~30 seconds, and the paper reports ~2.35 seconds per epoch on GPU for Cora. For graphs with millions of nodes, mini-batch training with graph sampling would be necessary, but the paper does not discuss or implement this. The covariance computation in Eq. (4) also requires that the hidden dimension p be smaller than the minimum node degree, otherwise regularisation (adding cI to the covariance) is needed — a practical constraint that limits the expressiveness of representations for low-degree nodes.

6. Summary

GAD-NR makes a strong conceptual contribution by identifying neighbourhood reconstruction as the right objective for comprehensive graph anomaly detection. However, the approach has meaningful limitations: (1) the 1-hop restriction misses higher-order anomalies; (2) the Gaussian assumption is violated in multi-community neighbourhoods; (3) the anomaly score weighting and loss scheduling are heuristic; (4) evaluation relies heavily on synthetic anomaly injection and a narrow set of graph types; and (5) scalability to very large or dynamic graphs is unaddressed. These issues present clear opportunities for improvement, which we explore in our proposed approach (Task 2).